National System Operator (Pvt) Ltd
Sri Lanka

POWER SYSTEM STRENGTHENING AND RENEWABLE ENERGY INTEGRATION PROJECT
(ADB LOAN NO: 4548-SRI)

PACKAGE 6

Procurement of Plant

Design, Supply, and Installation

Single-Stage: Two-Envelope
Bidding Procedure

BIDDING DOCUMENT
FOR

Integration of Renewable Energy Management System (REMS) and NSCC SCADA/EMS Modernization

Information Copy

VOLUME 7 of 8
Part II- REQUIREMENTS
Section 6 - Employer's Requirements: Part D-Supplementary Information, Part E-Bank Guaranties and Certificates, Change Orders

Issued on: March 30, 2026
Invitation for Bids No.: CEB/AGM(P)/TDE/IFB/2025/02/HLPC
OCB No.: CEB/AGM(P)/TDE/OCB/2025/02/HLPC
Employer: National System Operator (Pvt) Ltd
Country: Sri Lanka

Office of the Director (System Control)
National System Operator (Pvt) Ltd,
No. 80, Parliament Road,
Pelawatta, Battaramulla.
Sri Lanka.
Document – Revision 1

© Director Engineering, NTNSP

Information Copy

Preface

This Bidding Document for Procurement of Plant – Design, Supply, and Installation, has been prepared by National Transmission Network Service Provider (Private) Limited and is based on the Standard Bidding Document for Procurement of Plant – Design, Supply, and Installation (SBD Plant) issued by the Asian Development Bank dated December 2021.

ADB's SBD Plant has the structure and the provisions of the Master Procurement Document entitled "Procurement of Plant – Design, Supply, and Installation", prepared by multilateral development banks and other public international financial institutions except where ADB-specific considerations have required a change.

In this Bidding Document, any reference to 'Ceylon Electricity Board' shall be deemed to refer to 'National System Operator (Pvt) Ltd.' Similarly, any reference to 'CEB' shall be deemed to refer to 'NSO'.

Information Copy

Information Copy

Table of Contents - Summary Description

VOLUME 1 OF 8

PART I BIDDING PROCEDURES

Section 1 - Instructions to Bidders (ITB) ---------- **1-1**
This Section specifies the procedures to be followed by Bidders in the preparation and submission of their Bids. Information is also provided on the submission, opening, and evaluation of bids and on the award of contract.

Section 2 - Bid Data Sheet (BDS) ---------- **2-1**
This Section consists of provisions that are specific to each procurement and supplement the information or requirements included in Section 1 - Instructions to Bidders.

Section 3 - Evaluation and Qualification Criteria (EQC) ---------- **3-1**
This Section contains the bid evaluation criteria to determine the lowest evaluated bid and lists the necessary qualifications of Bidders.

VOLUME 2 OF 8

PART I BIDDING PROCEDURES

Section 4 - Bidding Forms (BDF) ---------- **4A-1**
Part A-Price Bid
This Section contains the forms which are to be completed by the Bidder and submitted as part of its Bid.

VOLUME 3 OF 8

PART I BIDDING PROCEDURES

Section 4 - Bidding Forms (BDF) ---------- **4B-1**
Part B-Technical Bid
This Section contains the forms which are to be completed by the Bidder and submitted as part of its Bid.

Section 5 - Eligible Countries (ELC) ---------- **5-1**
This Section contains the list of eligible countries.

VOLUME 4 OF 8

PART II REQUIREMENTS

Section 6 - Employer's Requirements (ERQ) ---------- **6A-1**
Part A-Scope of Works

VOLUME 5 OF 8

PART II REQUIREMENTS

Section 6 - Employer's Requirements (ERQ) ---------- **6B-1**
Part B-Technical Specification

Information Copy

VOLUME 6 OF 8

PART II REQUIREMENTS

Section 6 - Employer's Requirements (ERQ) ---------- **6C-1**
Part C-Drawings

VOLUME 7 OF 8

PART II REQUIREMENTS

Section 6 - Employer's Requirements (ERQ) ---------- **6D-1**
Part D-Supplementary Information
Section 6 - Employer's Requirements (ERQ) ---------- **6D-1**
Part E-Bank Guarantees and Certificates, Change Orders

VOLUME 8 OF 8

PART III CONDITIONS OF CONTRACT AND CONTRACT FORMS

Section 7 - General Conditions of Contract (GCC) ---------- **7-1**
This Section contains the general clauses to be applied in all contracts. These Conditions are subject to the variations and additions set out in Section 8 (Special Conditions of Contract).

Section 8 - Special Conditions of Contract (SCC) ---------- **8-1**
This Section contains provisions that are specific to each contract and that modify or supplement the GCC. Whenever there is a conflict, the provisions herein shall prevail over those in the GCC. The clause number of the SCC is the corresponding clause number of the GCC.

Section 9 - Contract Forms (COF) ---------- **9-1**
This Section contains forms, which, once completed, will form part of the Contract. The forms for Performance Security and Advance Payment Security, when required, shall only be completed by the successful Bidder after contract award.

Information Copy

Information Copy

Information Copy

Employer's Requirements

SUPPLEMENTARY INFORMATION

CONTENTS

1 MANUFACTURERS, PLACE OF MANUFACTURE AND TESTING 4
2 TECHNICAL PARTICULARS AND GURANTEES 5
A - ELECTRICAL WORKS 5
2.1 Enterprise Server 7
2.2 SCADA Server 9
2.3 SOC Server 11
2.4 FEP Server 13
2.5 OTS Server 15
2.6 Firewall 17
2.7 Control Centre Operational Display 19
2.8 Renewable Energy Management System (REMS) 21
2.9 Renewable Energy Forecasting System 23
2.10 Demand Forecasting System 25
2.11 Dispatch Scheduling and AGC System 27
2.12 Transparent Generation Management and Curtailment System 30
2.13 Transmission Network Security Assessment System 32
2.14 Battery Energy Storage System (BESS) Monitoring and Controlling 34
2.15 Real-time Operational Requirements Assessment 36
2.16 Power Plant and Network Outage Management System 38
2.17 Modernization of the SCADA/EMS Master Station 40
2.18 Cybersecurity Enforcement 42
2.19 Remote Station Adaptation Works 44
2.20 Ancillary Civil, Mechanical, Electrical, and Plumbing (MEP) Infrastructure 46
2.21 REMS Supporting Tools and Simulation Systems 48
2.22 Provisions for Market Integration and Cross-Border Interconnection 50
2.23 SCADA System 52
2.24 API-BASED COMMUNICATIONS FOR DER AND BESS 54
2.25 STUDY MODE NETWORK ANALYSIS 56
2.26 Reactive Power Scheduling 58
B - CIVIL WORKS 60

Information Copy

2.28 EXTERNAL LIGHTING AND SMALL POWER SUPPLY SERVICES 60
3 TIMES FOR DELIVERY & COMPLETION AND CONTRACT COMPLETION TIMES 63
3.1 TIMES FOR DELIVERY AND COMPLETION 63
3.2 CONTRACT COMPLETION TIMES 65
4 DEVIATIONS FROM SPECIFICATION 66
5 MANUFACTURES' AND SUBCONTRACTORS' STATEMENT OF EXPERIENCE 67
5.1 Tenderer's Statement of Previous Experience 67
5.2 Key Personnel 69
5.3 Contractor's Site Personnel 70
5.4 Sub-Contractors 71
6 DOCUMENTS, DRAWINGS AND INFORMATION TO BE SUBMITTED WITH THE TECHNICAL BID .. 72
7 ADHERENCE TO THE ENVIRONMENTAL ACTS, REGULATIONS AND / OR GUIDELINES 75

Information Copy

Notes on Schedules

The schedules are intended to provide the Employer with essential supplementary information in an organized format. The examples of more commonly used schedules are given herein. Others may be devised and added in accordance with the requirements of the instructions to bidders.

All the schedules are essential for the bid evaluation and some in contract execution; they should all be incorporated in the contract, and appropriate changes introduced with the approval of the employer or its representative.

The schedules are to be completed and submitted as part of the technical proposal in accordance with the instruction to bidders.

There shall be no equipment offered without filling these schedules. These sheets to be copied and filled in separately for each different types of equipment offered.

Information Copy

1 MANUFACTURERS, PLACE OF MANUFACTURE AND TESTING

Item	Manufacturer	Place of Manufacture	Place of Testing & Inspection
SCADA/EMS MASTER STATION SYSTEM SOFTWARE			
Operating Systems for Virtualized System			
Operating Systems for Virtual Machines/ Workstations			
Antivirus Software			
SCADA/EMS MASTER STATION HARDWARE			
Servers			
Workstations			
Control Centre Operational Display and Infrastructure			
Control Centre Operational Display			
System Displays			
Local Area Network (LAN) - Switches, Routers, & Cabling			
Storage Area Network (SAN)			
GPS Time Server			
SCADA/EMS MASTER STATION APPLICATIONS			
Scada Applications			
Energy Management System			
Supplementary Applications			
OTHER INFRASTRUCTURE			
Control Room Furniture			
All Additional Cubicles & Ancillary Equipment			
Cables			

Information Copy

2 TECHNICAL PARTICULARS AND GURANTEES

A - ELECTRICAL WORKS

Information Copy

Notes:

1. There shall be no equipment offered without filling these schedules. These sheets to be copied and filled in separately for each different type of equipment offered.

2. Refer scope of work & drawings for detail ratings.

Information Copy

2.1 ENTERPRISE SERVER

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Number of Physical Servers	No.	2		Configured with virtualization layer
Enterprise Servers	-	Server A/B		Redundant server pairs
Enterprise Server – Functions	-	Web, Patch, Historian interfaces		HTTPS only via reverse proxy
Enterprise Server DMZ – Security	-	Perimeter Firewall with IDS/IPS		Segregated from SCADA LAN
Operating System				
Processor Capacity	Cores	Min 16 per server		Dual CPU configuration
Memory	GB	Min 256 per server		Expandable to 512 GB
Local Storage	GB	Storage requirement For Hypervisor installation: Minimum 2 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 1 or Dual Redundant SD cards Minimum 32GB per card and For Local storage: Minimum 10 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 6 for local data		For hypervisor and config

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
		storage. Minimum usable capacity needs to be 8 TB.		
Virtual Machines per Cluster	-	Min 50 VMs		Covers SCADA, App, FEP, SOC, OTS, Historian
SAN Usable Capacity	TB	Min 30		Configured with RAID 6
SAN Redundancy	-	Dual Controller		Hot-swap, dual power supply
Backup Retention	Days	Min 90		Daily snapshots + DR replication
Time Synchronization Accuracy	ms	≤ 1		GPS/PTP-synchronized system
Role-based Access Control	-	Mandatory		With MFA & PAM
SIEM/SOAR Integration	-	Mandatory		Centralized alerting & correlation
Patch Management	-	Automated		Via DMZ Patch Servers
Compliance	-	IEC 62443 SL2–3, NIST CSF		Across all clusters
Testing & Acceptance	-	FAT, SAT, SIT		Full system validation
Operator & SOC Training	-	Mandatory		Including cybersecurity & OTS
Documentation	-	As-built drawings, configs, SOPs		Delivered upon completion
Warranty	Years	Min 3 years		Onsite with quarterly audits

Information Copy

2.2 SCADA SERVER

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Number of Physical Servers	No.	4		Configured with virtualization layer
Operations Server DMZ – Servers	-	SCADA A/B, App A/B		Redundant server pairs
Operations DMZ – Security	-	Redundant OT Firewalls with IDS/IPS		Strict allow-list rules
SCADA LAN – Servers	-	SCADA Server Cluster, Application Server Cluster, Historian		Full redundancy
SCADA LAN – Workstations	No.	Min 5 Operator Consoles		Multi-monitor, 24/7 operation
SCADA LAN – Security	-	RBAC + MFA, PAM vault		Internal Firewall protected
Operating System				
Processor Capacity	Cores	Min 16 per server		Dual CPU configuration
Memory	GB	Min 256 per server		Expandable to 512 GB
Local Storage	GB	Storage requirement For Hypervisor installation: Minimum 2 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 1 or Dual Redundant SD cards Minimum 32GB per card and		For hypervisor and config

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
		For Local storage: Minimum 10 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 6 for local data storage. Minimum usable capacity needs to be 8 TB.		
Virtual Machines per Cluster	-	Min 50 VMs		Covers SCADA, App, FEP, SOC, OTS, Historian
Backup Retention	Days	Min 90		Daily snapshots + DR replication
Time Synchronization Accuracy	ms	≤ 1		GPS/PTP-synchronized system
Role-based Access Control	-	Mandatory		With MFA & PAM
SIEM/SOAR Integration	-	Mandatory		Centralized alerting & correlation
Patch Management	-	Automated		Via DMZ Patch Servers
Compliance	-	IEC 62443 SL2–3, NIST CSF		Across all clusters
Testing & Acceptance	-	FAT, SAT, SIT		Full system validation
Operator & SOC Training	-	Mandatory		Including cybersecurity & OTS
Documentation	-	As-built drawings, configs, SOPs		Delivered upon completion
Warranty	Years	Min 3 years		Onsite with quarterly audits

Information Copy

2.3 SOC SERVER

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Number of Physical Servers	No.	2		Configured with virtualization layer
SOC Servers	-	Server A/B		Redundant server pairs
SOC LAN – Servers	-	Active Directory (OT), SIEM/SOAR, Patch Mgmt		Centralized governance
SOC LAN – Security	-	Log aggregation & monitoring only		No control functions
Operating System				
Processor Capacity	Cores	Min 16 per server		Dual CPU configuration
Memory	GB	Min 256 per server		Expandable to 512 GB
Local Storage	GB	Storage requirement For Hypervisor installation: Minimum 2 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 1 or Dual Redundant SD cards Minimum 32GB per card and For Local storage: Minimum 10 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid		For hypervisor and config

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
		6 for local data storage. Minimum usable capacity needs to be 8 TB.		
Virtual Machines per Cluster	-	Min 50 VMs		Covers SCADA, App, FEP, SOC, OTS, Historian
Backup Retention	Days	Min 90		Daily snapshots + DR replication
Time Synchronization Accuracy	ms	≤ 1		GPS/PTP-synchronized system
Role-based Access Control	-	Mandatory		With MFA & PAM
SIEM/SOAR Integration	-	Mandatory		Centralized alerting & correlation
Patch Management	-	Automated		Via DMZ Patch Servers
Compliance	-	IEC 62443 SL2–3, NIST CSF		Across all clusters
Testing & Acceptance	-	FAT, SAT, SIT		Full system validation
Operator & SOC Training	-	Mandatory		Including cybersecurity & OTS
Documentation	-	As-built drawings, configs, SOPs		Delivered upon completion
Warranty	Years	Min 3 years		Onsite with quarterly audits

Information Copy

2.4 FEP SERVER

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Number of Physical Servers	No.	2		Configured with virtualization layer
FEP Servers	-	Server A/B		Redundant server pairs
FEP LAN – Protocols	-	ICCP, IEC 101/104, DNP3, Modbus, IEC 61850		Legacy + future support
FEP LAN – Security	-	Front-End OT Firewalls with IDS/IPS		TLS/IPsec where supported
FEP LAN – Availability	-	Redundant FEP Servers		Hot-standby or active-active
Operating System				
Processor Capacity	Cores	Min 16 per server		Dual CPU configuration
Memory	GB	Min 256 per server		Expandable to 512 GB
Local Storage	GB	Storage requirement For Hypervisor installation: Minimum 2 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 1 or Dual Redundant SD cards Minimum 32GB per card and For Local storage: Minimum 10 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable		For hypervisor and config

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
		Drives in Raid 6 for local data storage. Minimum usable capacity needs to be 8 TB.		
Virtual Machines per Cluster	-	Min 50 VMs		Covers SCADA, App, FEP, SOC, OTS, Historian
Backup Retention	Days	Min 90		Daily snapshots + DR replication
Time Synchronization Accuracy	ms	≤ 1		GPS/PTP-synchronized system
Role-based Access Control	-	Mandatory		With MFA & PAM
SIEM/SOAR Integration	-	Mandatory		Centralized alerting & correlation
Patch Management	-	Automated		Via DMZ Patch Servers
Compliance	-	IEC 62443 SL2–3, NIST CSF		Across all clusters
Testing & Acceptance	-	FAT, SAT, SIT		Full system validation
Operator & SOC Training	-	Mandatory		Including cybersecurity & OTS
Documentation	-	As-built drawings, configs, SOPs		Delivered upon completion
Warranty	Years	Min 3 years		Onsite with quarterly audits

Information Copy

2.5 OTS SERVER

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Number of Physical Servers	No.	2		Configured with virtualization layer
FEP Servers	-	Server A/B		Redundant server pairs
FEP LAN – Protocols	-	ICCP, IEC 101/104, DNP3, Modbus, IEC 61850		Legacy + future support
FEP LAN – Security	-	Front-End OT Firewalls with IDS/IPS		TLS/IPsec where supported
FEP LAN – Availability	-	Redundant FEP Servers		Hot-standby or active-active
Operating System				
Processor Capacity	Cores	Min 16 per server		Dual CPU configuration
Memory	GB	Min 256 per server		Expandable to 512 GB
Local Storage	GB	Storage requirement For Hypervisor installation: Minimum 2 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable Drives in Raid 1 or Dual Redundant SD cards Minimum 32GB per card and For Local storage: Minimum 10 x 1TB, 2.5"/3.5", 12G SSD hot-pluggable		For hypervisor and config

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
		Drives in Raid 6 for local data storage. Minimum usable capacity needs to be 8 TB.		
Virtual Machines per Cluster	-	Min 50 VMs		Covers SCADA, App, FEP, SOC, OTS, Historian
Backup Retention	Days	Min 90		Daily snapshots + DR replication
Time Synchronization Accuracy	ms	≤ 1		GPS/PTP-synchronized system
Role-based Access Control	-	Mandatory		With MFA & PAM
SIEM/SOAR Integration	-	Mandatory		Centralized alerting & correlation
Patch Management	-	Automated		Via DMZ Patch Servers
Compliance	-	IEC 62443 SL2–3, NIST CSF		Across all clusters
Testing & Acceptance	-	FAT, SAT, SIT		Full system validation
Operator & SOC Training	-	Mandatory		Including cybersecurity & OTS
Documentation	-	As-built drawings, configs, SOPs		Delivered upon completion
Warranty	Years	Min 3 years		Onsite with quarterly audits

Information Copy

2.6 FIREWALL

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Firewall Type		Enterprise / Industrial Grade NGFW suitable for SCADA/EMS		Purpose-built for OT/ICS environments
Compliance Standard		IEC 62443		IEC 62443-3-3 and IEC 62443-4-2
Deployment Architecture		High Availability (HA)		Active-Active or Active-Standby
Failover Time	ms	≤ 1 second		No session loss
Firewall Throughput (IMIX)	Gbps	≥ 10		With all security features enabled
Maximum Latency	µs	≤ 100		Under full load
Concurrent Sessions	Number	≥ 5,000,000		Guaranteed minimum
New Connections Rate	connections/sec	≥ 200,000		With DPI enabled
VPN Throughput	Gbps	≥ 2		IPSec / SSL
SCADA Protocol Support		IEC 101, IEC 104, IEC 61850, ICCP		Native / DPI-based
Deep Packet Inspection		Mandatory		Protocol aware
RBAC		Mandatory		Granular roles
Multi-Factor Authentication		Mandatory		Privileged users
SOC / SIEM Integration		Mandatory		Syslog / SNMP / API
Time Synchronization		GPS / NTP		Aligned with Master Station
Power Supply Redundancy		Dual redundant		Hot-swappable
Mounting Type		Rack-mounted		Data-centre compatible
Operating Temperature	°C	0 to 45		Control-room environment

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Licenses & Subscriptions	Years	Minimum 5		Inclusive
Documentation & Training		Mandatory		As-built & training

Information Copy

2.7 CONTROL CENTRE OPERATIONAL DISPLAY

No.	Particular	Unit	Guaranteed Requirement	Offered Value (by Bidder)	Remarks
1.1	Video Wall Technology	-	Direct View LED (preferred)		
1.2	Operation	-	24 × 7 continuous mission-critical use		
1.3	Wall Dimensions	m	≥ 10.0 (W) × 2.25 (H) (adjustable as per layout)		
2.1	Pixel Pitch	mm	≤ 1.2		
2.2	Native Resolution (per module)	pixels	≥ Full HD (1920 × 1080)		
2.3	Overall Resolution (wall)	-	≥ 8K horizontal equivalent		
2.4	Brightness	nits	≥ 600 (auto adjustable with ambient sensor)		
2.5	Contrast Ratio	-	≥ 4000:1		
2.6	Viewing Angle	°	≥ 160° (H/V)		
2.7	Color Uniformity	ΔE	≤ 2		
2.8	Refresh Rate	Hz	≥ 60 (preferred ≥ 120)		
2.9	Lifetime (to 50% brightness)	hrs	≥ 100,000		
2.10	Bezel / Gap	mm	Seamless (zero visible bezel)		
3.1	Simultaneous Display Windows	-	≥ 32, scalable		
3.2	Supported Inputs	-	HDMI, DisplayPort, DVI, VGA, SDI, IP Streams (H.264/H.265), CCTV		
3.3	Controller Redundancy	-	Dual Ethernet, redundant power, hot-swappable fans		
3.4	Operator GUI	-	Drag-and-drop, resize, overlap, preset layouts		
3.5	Preset Layouts	-	Normal, Contingency, Emergency, Maintenance		
4.1	Maintenance Access	-	Front or Rear service access		
4.2	Wall Elevation	m	Raised ≥ 1.5 above finished floor		
4.3	Enclosures	-	Side panels and finishing to Employer's approval		
4.4	Seismic Compliance	-	Frame & mounting structure designed to withstand seismic vibrations		

Information Copy

No.	Particular	Unit	Guaranteed Requirement	Offered Value (by Bidder)	Remarks
5.1	MTBF (system)	hrs	≥ 50,000		
5.2	Power Supplies	-	Redundant, hot-swappable		
5.3	Cooling	-	Redundant, hot-swappable fans		
6.1	SCADA/EMS/REMS Integration	-	Native IP & workstation output integration		
6.2	Access Control	-	Role-based (operator/supervisor/admin)		
6.3	Remote Support	-	Remote diagnostics & firmware upgrade capability		
7.1	Warranty	years	≥ 5 (comprehensive)		
7.2	Local Support	hrs	24 × 7 availability, ≤ 4-hour response		
7.3	Training	-	Operator & Administrator training to be provided		

Information Copy

2.8 RENEWABLE ENERGY MANAGEMENT SYSTEM (REMS)

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Real-Time Plant Monitoring Capability	-	≥ 100 RE plants		Solar, Wind, Mini-Hydro
Forecast Horizons	-	Day-ahead, Intra-day, Nowcasting		For all RE sources
Forecast Model Accuracy	% (MAE)	≤ 5%		System-wide aggregate
Forecast Model Types	-	Statistical + ML Hybrid		NWP, telemetry, satellite inputs
Telemetry Protocol Support	-	IEC 61850, Modbus, IEC 104		Plant controller integration
Curtailment Logic Engine	-	Rule-based (Pro-rata, Priority)		With event log and timestamp
Reactive Power Monitoring	-	Supported		Per site with VAR visibility
Ride-through Compliance Monitoring	-	Voltage & Frequency		Inverter-level profiling
System Availability	%	≥ 99.5		Annual operational uptime
Control Dashboard Latency	Seconds	≤ 5		Signal-to-display update time
Data Historian Retention	Years	≥ 5		Operational + Forecast data
Report Generation Formats	-	PDF, Excel, XML		Exportable via dashboard
Regulatory Reporting Templates	-	Included		Daily, Monthly, Annual
Time Synchronization	-	NTP / GPS		For telemetry and alarms
Cybersecurity Standards	-	IEC 62351, NIST SP 800-82		Role-based access, MFA, logging

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Interoperability with SCADA/EMS	-	Bidirectional		Telemetry, control, and forecast sharing
API Access for 3rd Parties	-	REST/JSON or OPC UA		Market or analytics integration
DER and Rooftop Solar Ready	-	Yes		Scalable for future needs
FAT/SAT Compliance	-	Mandatory		With multi-source data validation
Training & Documentation	-	Operator + Admin		Manuals, SOPs, Interface control

Information Copy

2.9 RENEWABLE ENERGY FORECASTING SYSTEM

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Forecast Horizon (Day-Ahead)	Hours	≥ 48		Hourly intervals
Forecast Horizon (Intra-Day)	Hours	≥ 6		15-minute intervals
Forecast Horizon (Nowcasting)	Minutes	≥ 60		5-minute intervals
Forecast Resolution	-	Plant, Zonal, System-wide		Granular spatial support
Forecast Accuracy (MAE/MAPE)	%	≤ 5%		System-wide hourly average
Forecast Update Frequency	Minutes	≤ 15		For all active forecast levels
Input Data Sources	-	NWP, satellite, telemetry		Integrated weather and plant data
Forecast Methodology	-	Hybrid (Statistical + ML)		AI-enabled accuracy improvement
Forecast Performance Logging	-	MAE, RMSE, MAPE		Accuracy metrics stored
Visualization Latency	Seconds	≤ 30		Display on dashboard
Dashboard Access	-	Web & Desktop		Operator and analyst view
Integration with SCADA/EMS	-	Yes		For dispatch and curtailment linkage
API/Data Format Compatibility	-	REST API, CSV, XML, JSON		External access supported
System Availability	%	≥ 99.5%		On annual basis
Disaster Recovery Capability	-	Yes		With redundancy and backups
Forecast Archive Retention	Years	≥ 5		For accuracy and audit review

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Model Calibration & Retraining	-	Yes		Tool-based, ongoing learning
Acceptance of accuracy		System-wide MAPE ≤ 5% day-ahead (hourly), and ≤ 3% intra-day (15-minute) during steady-weather windows; computed on a rolling 30-day basis excluding declared curtailment hours.		

Information Copy

2.10 DEMAND FORECASTING SYSTEM

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Forecast Granularity (Short-Term)	Minutes	5, 15, 60		User-selectable resolution
Short-Term Forecast Horizon	Hours	≥ 48		5–60 minute intervals
Medium-Term Forecast Horizon	Days	≥ 30		Daily granularity
Long-Term Forecast Support	Months/Years	Optional		Monthly/annual trend projection
Forecast Resolution	-	Substation / Feeder / Regional		Multi-level (national, regional, feeder)
Forecast Accuracy (Short-Term)	% MAPE	≤ 5%		15-minute/1-hour intervals
Forecast Update Frequency	Minutes	≤ 15		Forecast refresh cycle
Forecast Input Integration	-	Historical, Weather, Events		Multi-dimensional input support
Forecast Methodology	-	ARIMA + ML Hybrid		Regression, ensemble, neural networks
Forecast Scenario Simulation	-	Supported		Custom load/demand patterns
System Availability	%	≥ 99.5%		High-availability infrastructure
SCADA/EMS Integration	-	Mandatory		Real-time dispatch and reserve interface
API/Data Format Support	-	JSON, CSV, XML		For external tools and cloud services

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
User Dashboard with Trend Visuals	-	Required		Historical trends, deviation alerts
Time Synchronization	-	NTP / GPS		Forecast timestamps for dispatch integration
Cybersecurity Compliance	-	Role-Based Access, Logging		OT/IT security standards
Data Retention	Years	≥ 5		Historical forecasts and accuracy data
Model Retraining & Calibration	-	Supported		Continuous learning with new data
Deployment Flexibility	-	On-Premise / Cloud		Supports both architectures
Disaster Recovery and Backup	-	Mandatory		Data and model redundancy
Implementation Testing	-	FAT + SAT		Accuracy and integration validation

Information Copy

2.11 DISPATCH SCHEDULING AND AGC SYSTEM

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
OMS-approved outages		Automatically modify unit participation in AGC		Required
Short-Term Scheduling Horizon	Hours	≥ 24		Updated hourly
Day-Ahead Scheduling Horizon	Hours	≥ 48		Must use forecast data
Intra-Day Rescheduling	-	Yes		Triggered by real-time deviations
Forecast-Based Reserve Calculation	-	Supported		Based on RE + Load uncertainty
AGC Control Modes	-	Flat frequency, Tie-line bias, Coordinated		Must support all modes
AGC Response Time	Seconds	≤ 4		Setpoint signal to generator response
AGC Signal Resolution	MW	0.1		Granularity of command signal
AGC-Enabled Generator Count	Units	≥ 10		Scalable with grid needs
Setpoint Dispatch Frequency	Seconds	≤ 10		Continuous control cycle
Secure delivery of Start/Stop commands to eligible plants				Required
All setpoints and Start/Stop commands shall use the unified SCADA/EMS command lifecycle				Required
The system shall record full command traceability including origin,				Required

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
timestamp, payload, acknowledgement, and plant response				
Control Accuracy	%	≥ 95%		ACE or frequency control accuracy
System Availability	%	≥ 99.98%		Real-time control and monitoring
Control Signal Redundancy	-	Dual-path		Resilient architecture
SCADA/EMS Integration	-	Mandatory		Real-time data exchange
Time Synchronization	-	GPS or NTP		For telemetry and control timestamping
Communication Protocols	-	IEC 60870-5-101, IEC 60870-5-104, IEC 61850, and secure API-based		Standards-based only
Alarm & Deviation Alerts	-	Automatic Non-responsive units, AGC blocking conditions, and abnormal deviations		Integrated with operator console
Historical Reporting	-	Daily & Monthly		For audits and regulatory use
Data Retention	Years	≥ 5		AGC response and scheduling history
Operator Training Simulator	-	Required		Scenario-based simulation
Role-based access control		Enforced for AGC operation and overrides		Required

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Future Integration Readiness	-	AI/Market modules		Interface-ready architecture
Real-time displays		ACE, unit MW outputs, setpoints, and participation factors		Required

Information Copy

2.12 TRANSPARENT GENERATION MANAGEMENT AND CURTAILMENT SYSTEM

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Curtailment Logic Modes	-	Pro-rata, LIFO, Grid Security		User-configurable, rules-based
Curtailment Control Modes	-	Manual & Automated		Switchable via operator interface
Timestamp Resolution for Events	ms	≤ 100		GPS/NTP synchronized
Response Time (Command to Acknowledgment)	Seconds	≤ 3		Control signal round-trip confirmation
System Availability	%	≥ 99.5		Annual uptime for UI and control
Integration Interfaces	-	SCADA/EMS, REMS		Via IEC 104 / 61850 / Modbus TCP
Real-Time Visualization	-	Required		With location, magnitude, and duration
Historical Trend Depth	Years	≥ 5		Archive with filtering options
Audit Trail Completeness	-	Yes		All curtailments logged with metadata
Report Generation Formats	-	PDF, Excel, XML		Configurable and exportable
External Stakeholder Dashboard	-	Optional		Read-only web access for IPPs, regulator
Alarm/Event Notification	-	Yes		Triggered on unexpected deviations
Data Retention Duration	Years	≥ 5		Includes logs, signals, and actions
APIs for External Access	-	REST or Web-based		With authentication
User Access Control	-	RBAC with Logging		Multi-tier permission model

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
FAT and SAT Compliance	-	Mandatory		Curtailment logic and report validation
Scalability to DERs/Microgrids	-	Supported		Expandable architecture
Analytics Features	-	Root Cause, Forecasting (optional)		Optional AI/ML modules supported

Information Copy

2.13 TRANSMISSION NETWORK SECURITY ASSESSMENT SYSTEM

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Contingency Simulation Support	-	N-1 (Mandatory), N-2 (Optional)		Automated ranking and visualization
Contingency Analysis Refresh Rate	Minutes	≤ 10		Dynamic update with real-time inputs
Supported Input Sources	-	SCADA, State Estimator, PMUs		Via secure, time-synced interface
Voltage and VAR Assessment	-	Yes		With compensation device modeling
Predictive Security Forecasting	-	Yes		Using RE and load forecast integration
Visual Dashboard with Risk Index	-	Mandatory		With color-coded risk indicators
Real-time Constraint Alarm Latency	Seconds	≤ 30		Operator alert generation
What-if Scenario Simulation	-	Yes		For switching, generation, and VAR control
Security Margin Index Output	-	Yes		For operator decisions and logging
Interoperability with PSS®E / PowerFactory	-	Yes		For offline analysis compatibility
Time Synchronization Protocol	-	GPS / NTP		For SOE accuracy and timestamp matching
System Availability	%	≥ 99.5%		Over a 12-month operational window
Historical Data Retention	Years	≥ 5		Security assessment and

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
				intervention logs
Integration with SCADA/EMS	-	Mandatory		For network model and real-time telemetry
Cybersecurity Compliance	-	IEC 62351, ISO 27001		Access, logging, and secure storage
Training Simulator for Operator Scenarios	-	Included		For NSCC capacity building
Disaster Recovery and Backup	-	Yes		Daily sync with secure off-site storage
Future AI/ML Readiness	-	Recommended		Security index prediction capability

Information Copy

2.14 BATTERY ENERGY STORAGE SYSTEM (BESS) MONITORING AND CONTROLLING

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Monitoring Resolution	-	Real-time + Historical		Including SoC, SoH, voltage, temp, alarms
Data Update Frequency	Seconds	≤ 10		System-wide telemetry refresh
Round-Trip Efficiency Reporting	%	Supported		Cycle-level and daily reports
Degradation Trend Analysis	-	Yes		With historical model tracking
Thermal Profile Monitoring	-	Yes		Per battery module or string
Predictive Maintenance Analytics	-	Included		Cycle count, usage forecast, alarm correlation
Control Modes	-	Manual + Automatic		For charging, discharging, grid services
Dispatch Integration Capability	-	Yes		Receives commands from SCADA/EMS
AGC Response Support	ms	≤ 250		Setpoint following latency
Inverter/BMS Protocols Supported	-	Modbus TCP/IP, IEC 61850		Vendor-interoperable
Centralized Dashboard Availability	%	≥ 99.5		With geo-located multi-asset view
Alarm Escalation and Logging	-	Yes		Tiered alarm handling to NSCC
System Redundancy	-	Yes		Data storage, control, and communication
Cybersecurity Compliance	-	IEC 62443, NIST SP 800-82		RBAC, MFA, event logging
Historical Data Retention	Years	≥ 5		High-frequency storage with backup

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Data Export Format	-	PDF, Excel, XML		Daily/Monthly/On-demand reports
Interface with Fire & HVAC	-	Supported		Status monitoring and alarm relay
System Scalability	-	Yes		Multi-site, future DER fleet capability
FAT/SAT Requirement	-	Mandatory		Validation at factory and each BESS site
Future grid-forming support	-	Yes		Where supported, BESS shall expose parameters for grid-forming or grid-following mode, with droop settings, virtual inertia constants, and fast frequency response ≤250 ms. Control setpoints via IEC 61850 logical nodes (DER/ES).

Information Copy

2.15 REAL-TIME OPERATIONAL REQUIREMENTS ASSESSMENT

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Assessment Update Frequency	Minutes	≤ 5		Refresh cycle of operational indices
System Inputs Supported	-	SCADA, State Estimator, PMUs		Data ingestion capability
Voltage Stability Assessment	-	Yes		Simulated VAR needs and support devices
Reserve Margin Calculation	-	Yes		Based on RE forecast and system inertia
Congestion Detection & Ranking	-	Included		Line loading, forecast-based overloads
Critical Bus Voltage Identification	-	Yes		Var support recommendation included
Redispatch Simulation Support	-	Yes		Operator what-if analysis tool
Mitigation Action Recommendations	-	Yes		For voltage, reserve, and congestion issues
User Dashboard with Severity Ranking	-	Required		Categorized active operational needs
SCADA/EMS/REMS Integration	-	Mandatory		For telemetry, control, and forecasts
Event-Based Alarm Generation	-	Yes		For reserve shortfall or line constraints
Historical Log Retention	Years	≥ 5		Logged operator actions and events
Computation Engine Performance	-	High-speed		For real-time metrics and ranking
Time Synchronization	-	GPS / NTP		Timestamps aligned with SCADA/PMU

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Web and Thick Client Interface	-	Required		For planners and operator stations
Cybersecurity Compliance	-	IEC 62443, ISO 27001		RBAC, encryption, audit logs
Operator Training Support	-	Included		Use-case scenarios for practice
FAT/SAT Requirements	-	Mandatory		With validation of computation logic
AI/DER Data Compatibility	-	Recommended		Future integration with WAMS, DER, EV

Information Copy

2.16 POWER PLANT AND NETWORK OUTAGE MANAGEMENT SYSTEM

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Outage Request Submission Interface	-	Digital workflow		With calendar and PTW info
Approval Workflow Levels	Stages	≥ 3		Validation, technical, and operational
Outage Visualization Interface	-	Interactive calendar		Shows active/pending/completed
Outage Confirmation Latency	Seconds	≤ 60		From telemetry confirmation to status update
Real-Time Status Sync	-	SCADA-integrated		Breaker/gen signals auto-update
Unplanned Outage Classification	-	Mandatory		Auto-logging with root cause fields
Impact Simulation	-	Flow, voltage, reserve		Pre-outage modeling support
Outage Report Generation	-	Automatic		Post-event summary with deviations
Data Retention	Years	≥ 5		Outage logs, reports, annotations
Outage Summary Reporting	-	Monthly and Annual		By type, zone, asset
External System Integration	-	GIS, Scheduling, TNSA		Via API/REST interface
User Access Control	-	Role-Based		Operators, planners, regulators
Historical Analytics	-	Supported		For planning and risk profiling
Cybersecurity Compliance	-	IEC 62443 / ISO 27001		Access logging and encryption
Mobile Approval Option	-	Future-ready		Provision for app-based interface
Disaster Recovery and Backup	-	Yes		Redundant outage data archival

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Operator Training Module	-	Included		Outage process walkthroughs
FAT/SAT Verification	-	Mandatory		Workflow + monitoring validation

Information Copy

2.17 MODERNIZATION OF THE SCADA/EMS MASTER STATION

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Real-Time Data Point Capacity	Points	≥ 50,000		Scalable with future grid growth
Communication Protocol Support	-	IEC 60870-5-104, IEC 101, IEC 61850		Including backward compatibility
Historian Data Retention	Years	≥ 5		Operational data, alarms, and events
Operator Console Refresh Rate	Seconds	≤ 1		Sub-second update rate for HMI
Redundancy Type	-	Hot-Standby or Active-Active		For servers, FEPs, LAN, DB
Failover Switching Time	Seconds	≤ 30		No loss of critical data
Topology and State Estimation	-	Supported		Automatic or on-demand
Contingency Analysis	-	N-1 and configurable		With operator impact view
AGC Integration	-	Required		Load-frequency control with RE tools
Alarm/Event Handling	-	Dynamic, Rule-based		Graphical and time-stamped
Time Synchronization	-	GPS / NTP		For SOE and log precision
SCADA-Historian Interface	-	CSV, XML, SQL Export		Third-party platform ready
Cybersecurity Standards	-	IEC 62443, ISO 27001		MFA, RBAC, Patch Management
External Interface Capability	-	REMS, Forecasting, Outage Mgt.		Interoperable API/Protocol
Deployment Architecture	-	Modular, Distributed		Supports future upgrades
Cloud/Virtualization Ready	-	Yes		For backup and scalability
Market/IPP Integration Readiness	-	Yes		Cross-border and market-ready

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Operator Training & Simulation	-	Included		For scenario-based practice
FAT/SAT Compliance	-	Mandatory		Full functional validation

Information Copy

2.18 CYBERSECURITY ENFORCEMENT

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Compliance Standards	-	IEC 62443, ISO/IEC 27001, NIST SP 800-82		Must be met end-to-end
Identity and Access Control	-	RBAC + MFA		With full audit logging
Endpoint Hardening	-	Mandatory		Anti-malware, HIDS, secure configs
Patch Management Process	-	Automated + Auditable		All systems regularly updated
Log Retention Duration	Years	≥ 1		For audit and forensic analysis
SIEM Integration	-	Mandatory		Centralized logging, alerts, response
Incident Response Plan (IRP)	-	Required		Integrated with monitoring tools
Network Segmentation	-	IT/OT Zoning + DMZs		With ACLs and traffic monitoring
Protocol Security	-	VPN, TLS, SSH		For remote/data interfaces
External Threat Feed Support	-	Yes		Integrates with national CERT/SOC
System Availability	%	≥ 99.9		Across all monitoring and alerting subsystems
Configuration Backup & Recovery	-	Enabled		Full-system rollback on failure
Asset Inventory Management	-	Mandatory		Dynamic classification by risk
Audit Logging Coverage	-	Complete		Logins, changes, access, updates
Forensic Investigation Readiness	-	Yes		Indexed, exportable evidence store
Cybersecurity Training Coverage	-	All roles		Operators, admins, incident team
FAT/SAT Validation	-	Required		Functional and compliance testing

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
AI/Anomaly Detection Integration	-	Recommended		Optional upgrade path
Secure Development Practices	-	Mandatory		For all integrations/customizations
Remote Access Hardening	-	Firewalled + Authenticated		No default admin ports

Information Copy

2.19 REMOTE STATION ADAPTATION WORKS

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
RTU Protocol Compatibility	-	IEC 104, IEC 61850, Modbus		Native or via protocol converters
Minimum Data Points per Station	points	≥ 500		Analog, digital, status, alarms
SOE Timestamp Resolution	ms	≤ 1		Synchronized with GPS/NTP
Redundant Communication Paths	-	Dual Link		Fiber/MPLS/VPN with failover
Cybersecurity Compliance	-	RBAC, Secure Boot, Firmware Auth.		IEC 62443-based
Remote Firmware Upgradability	-	Yes		With version tracking and rollback
Unified Signal Naming & Scaling	-	Required		Standardized across all RTUs
Alarm Threshold Configuration	-	Centralized		NSCC-level control
RTU Environmental Rating	-	Industrial Grade		EMC, -10°C to +55°C, humidity ≤ 95%
RTU Diagnostic Features	-	Self-test, health monitoring		Accessible via secure GUI
Interlocking/Protection Compatibility	-	No disruption		Validated during SAT
Real-time Control Support	-	Breaker/Device Operations		With command feedback
RTU/Comm Testing Requirement	-	FAT + SAT		Including signal loop & failure simulation
Configuration Backup Format	-	XML/CSV + binary		Encrypted preferred
Data Integrity Verification	-	Checksum or heartbeat		System polling must confirm integrity
Substation Documentation Handover	-	As-built + Red-line		Schematics, configuration, wiring
Training to NSCC/Substation Staff	-	Mandatory		Systems, reset, alarms, signal maps

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Scalability to SAS/DER Integration	-	Yes		Future IEC 61850 asset support
Implementation Coordination	-	PTW + Station Lead		Site-specific execution plan

Information Copy

2.20 ANCILLARY CIVIL, MECHANICAL, ELECTRICAL, AND PLUMBING (MEP) INFRASTRUCTURE

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Control Room Design Compliance	-	24/7 vibration-free, fire-rated		Sound and thermal insulation
Raised Flooring	-	Installed		For cable routing and ventilation
PAC Units Redundancy	-	N+1		Per cooling zone
Temperature Range in IT Zones	°C	22 ± 2		Maintained via PAC
Humidity Range	% RH	50 ± 10%		Real-time BMS monitored
Fire Suppression in Sensitive Areas	-	FM200 / Novec 1230		Integrated with BMS
General Area Fire Protection	-	Sprinklers / Hydrants		Per fire safety code
UPS Backup Time	Minutes	≥ 30		Dual redundant UPS
Diesel Generator Backup	Hours	≥ 24		Auto-start enabled
Electrical Cabling	-	FRLS, labeled, tray-routed		With surge protection
MDB and Earthing	-	Dedicated per zone		Compliant with safety code
Restroom & Pantry Facilities	-	Included		Pressurized plumbing
Water Supply Segregation	-	Potable vs Fire		Backflow prevention included
Moisture Sensors in Equipment Rooms	-	Included		Drainage with alarms
CCTV Surveillance	-	24x7 Monitoring		With recording and analytics
Access Control	-	Biometric / RFID		With logging system

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Perimeter Intrusion Detection	-	Included		Alerts via centralized panel
Emergency Lighting	-	Installed		Auto-on during power failure
Compliance to Safety Code	-	Mandatory		National building, electrical, fire
Future Expansion Provisions	-	PAC, Generator, Rack, Conduit		Tier II/III design standards

Information Copy

2.21 REMS SUPPORTING TOOLS AND SIMULATION SYSTEMS

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Forecast Accuracy Analysis Metrics	-	MAE, RMSE, MAPE		By plant, zone, technology
Forecast Accuracy Visualization Delay	Seconds	≤ 30		Post-run analytics visibility
Forecast Performance Logging Retention	Years	≥ 5		Archived for model comparison
Simulation Scenario Types	-	RE ramping, curtailment, instability		Customizable input sets
Scenario Parameter Customization	-	Yes		User-defined test configuration
Telemetry Simulation Support	-	Yes		Injected into test model logic
Operator Training Simulator (OTS)	-	Real-time REMS replica		Training + assessment tools
Training Scenario Complexity	-	Frequency/voltage/grid events		Replay, evaluate, annotate
Automated Report Generation	-	Daily, Weekly, Monthly		Generation, accuracy, curtailment
Export Formats for Reports	-	PDF, Excel, XML		Schedule or on-demand export
Regulatory Format Compliance	-	PUCSL Ready		Supports Sri Lankan grid standards
Data Quality Dashboard	-	Required		Signal loss, latency, anomaly detection
Model Validation Sandbox	-	Yes		Pre-deployment test environment
API Availability	-	RESTful		External tool integration

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Security Isolation	-	Live vs Test Environments		Access and data segregation
Time Synchronization Support	-	NTP/GPS		Cross-system event correlation
Audit Logging & Version Control	-	Enabled		Tracked simulation/model changes
FAT and SAT Compliance	-	Required		Forecast and training tools validated
Trainer and Analyst Training	-	Included		System tools and replay functionality
Future ML/DER Support	-	Recommended		Forecasting + DER aggregation readiness

Information Copy

2.22 PROVISIONS FOR MARKET INTEGRATION AND CROSS-BORDER INTERCONNECTION

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Market Interfaces Supported	-	DAM, RTM, Ancillary		With nomination, dispatch, settlement
Market Data Exchange Capability	-	Bids, Schedules, Prices		API/Data Bus enabled
Dispatch Stack Visualization	-	Included		Merit-order, unit commitment
Curtailment Categorization Logic	-	Supported		By market participation class
Financial Settlement Interface	-	Metering, Billing, Reconciliation		Post-dispatch alignment
Tie-Line Monitoring Capability	-	Power, Voltage, Frequency		Bi-directional exchange
Tie-Line Control Strategies	-	TBC, ACE, Frequency Control		For multi-area operation
Data Exchange Protocol	-	ICCP/TASE.2, IEC 61970/61968		Cross-control center links
Communication Security	-	VPN, HTTPS, SFTP		Encrypted with certificates
Control Center Link Redundancy	-	Yes		For regional grid integration
Cross-Border Forecast Visualization	-	Supported		Import/export trends, congestion
Contract Deviation Alarm	-	Enabled		Mismatch alerts and logging
Historical Archive Retention	Years	≥ 5		Cross-border dispatch and events
Middleware/ESB Integration	-	Yes		Message routing + protocol mediation

Information Copy

Parameter	Unit	Minimum Requirement	Guaranteed Value	Remarks
Cybersecurity Compliance	-	IEC 62443, NIST, ISO 27001		Full audit and secure access
Role-Based Access and Logging	-	Enabled		All market data transactions logged
Implementation Phasing Plan	-	Roadmap Required		Gap assessment and milestone planning
Simulation Environment for Testing	-	Required		Pre-deployment validation
Future Participant Scalability	-	Yes		Multi-market / multi-country ready
Regulatory Reporting Automation	-	Included		Aligned with PUCSL + regional markets

Information Copy

2.23 SCADA SYSTEM

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
SCADA System Availability	%	≥ 99.98%		Across all critical components
Main Control Server Redundancy	-	Hot-Standby / Active-Active		Failover without data loss
SCADA Data Point Capacity	points	≥ 100,000		Expandable for future substations
Telemetry Refresh Rate	seconds	≤ 2		Real-time updates to HMI
SOE Timestamp Resolution	ms	1 ms		With GPS-synchronized clocks
Front-End Processor (FEP) Redundancy	-	1+1 Minimum		Dual redundant architecture
Alarm Latency	seconds	≤ 1		From field event to SCADA display
Control Command Execution Delay	seconds	≤ 2		Round trip with feedback
Supported Protocols	-	IEC 60870-5-104, 101, IEC 61850		Including legacy support via gateway
SCADA-Historian Archival Duration	years	≥ 5		With export and search functions
Concurrent Operator Workstations	no.	≥ 10		Scalable with future control zones
Graphical Interface Availability	%	≥ 99.5%		Web/desktop HMI availability
Cybersecurity Compliance	-	IEC 62443, ISO 27001		With MFA, RBAC, and audit logs
External System	-	ICCP/TASE.2, OPC UA		For REMS, EMS, Market, DER systems

Information Copy

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
Interface Capability				
Disaster Recovery Time (RTO)	minutes	≤ 5		Switch to Backup Control Centre
Data Loss Tolerance (RPO)	minutes	≤ 1		From backup sync latency
Power Backup Duration	minutes	≥ 30		For all control, FEP and server racks

Information Copy

2.24 API-BASED COMMUNICATIONS FOR DER AND BESS

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
API System		HTTPS-based RESTful APIs as the primary interface for DER/BESS data acquisition and control. Detect schema changes or abnormal data behaviour		Required
Optional support		SOAP/XML, IEC 61850-7-420, IEEE 2030.5, and OpenADR 2.0b		Required
API subsystem		Acquire telemetry, status, alarms, forecasts, and SOC/SOH for BESS		Required
Middleware		Map third-party API schemas and control models. Support HA with load balancing and retries		Required
API communication		Polling and event-driven modes		Required
Typical telemetry update	≤ 5 Seconds			Required
All endpoint state changes and operator actions		Shall be logged and support OAuth2.0/token-based authentication and RBAC		Required

Information Copy

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
Loss of API communication		Raise alarms and retain last good value		Required
API metrics		Trended and exportable		Required

Information Copy

2.25 STUDY MODE NETWORK ANALYSIS

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
Study Mode		Execution of analytical applications identical to real-time (TP, SE, Load Flow, CA, OPF, Bus Forecast, Reactive Analysis, Loss Factors)		Operators shall retrieve, edit, and modify study cases using one-line diagrams and structured forms. Support Real-Time save cases created from the current SE-adjusted network state. Support Past save cases generated from RTDB, historian, events, measurements, and topology archives
Future save cases		Built using generation schedules from Dispatch Scheduling/AGC and approved outages from OMS		Include topology, load levels, generation outputs, DER forecasts, BESS state, and switching statuses.
Playback function		Application shall replay past system conditions (values, statuses, events)		System shall allow recreation of incident-specific save cases from any selected timestamp. Support time-synchronized viewing of measurements, statuses, DER outputs, switching actions, and

Information Copy

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
				frequency/voltage events.
Study sessions		At least three concurrent independent study sessions		Clear indicators shall display Study Mode vs Real-Time mode.
State Estimation		Usable to tune, initialize, and validate study models		Required
Save case retrieval	≥ 1 year	Support long-term archival and retrieval of historical save cases		Required

Information Copy

2.26 REACTIVE POWER SCHEDULING

Parameter	Unit	Minimum Requirement	Offered Value	Remarks
Reactive Power Scheduling System		Maintains secure voltage profiles and reactive balance in real-time and forecasted operation		Required
Reactive Power Scheduler		Supports synchronous generators, STATCOM/SVC/VSR, shunts, OLTCs, DER inverters, and BESS. Integrates with SE, AGC, REMS, DER forecasting, BESS management, and OMS outage data		Respects generator P–Q limits, DER VAR envelopes, BESS VAR limits, and device constraints
Reactive Power Scheduling Outputs		MVAr setpoints, DER VAR commands, BESS VAR setpoints, STATCOM/SVC dispatch, OLTC taps		Required
VAR setpoints		Issued automatically through SCADA/API using unified command lifecycle		Required
Optimization and Diagnostics		Runs at configurable intervals or under voltage/topology-triggered events. Identify binding constraints, infeasible scenarios, and weak buses		Required
System scalability		Future DER, BESS, STATCOM, VSR, and 400 kV network expansion		Required

Information Copy

Information Copy

2 TECHNICAL PARTICULARS AND GURANTEES

B - CIVIL WORKS

2.28 EXTERNAL LIGHTING AND SMALL POWER SUPPLY SERVICES

No	Item	Units	Required	Tendered
(a)	Distribution Boards (fitted with fuses)			
	Manufacturer's name and address			
	Type and/or Figure No.			
	Rating	A		
	Fault rating	kA		
	Voltage	V		
(b)	Distribution Boards (fitted with Circuit breakers)			
	Manufacturer's name and address			
	Type and/or Figure No.			
	Rating	A		
	Fault rating	kA		
	Voltage	V		
(c)	PVC Cable			
	Manufacturer's name and address			
	Type			
	Voltage rating	V		
(d)	Conduit			
	Manufacturer's name and address			
	Type			
(e)	Conduit Accessories			
	Manufacturer's name and address			
	Type			
(f)	Cable Termination's			
	Manufacturer's name and address			
	Type			

Information Copy

No	Item	Units	Required	Tendered
	Material			
(g)	Switches			
	Manufacturer's name and address			
	Type			
	Rating	W		
(h)	Socket Outlets			
	Manufacturer's name and address			
	Type and/or Figure No.			
	Rating	W		
	Finish			
(i)	Contactors			
	Manufacturer's name and address			
	Type			
	Rating	W		
	Number of contacts			
	Rating of coil AC	VA		
(j)	Miniature Circuit Breakers			
	Manufacturer's name and address			
	Type			
	Rating	A		
	Fault rating	kA		
(k)	Earthing Material			
	Manufacturer's name and address			
	Material & size			
(l)	Clocks			
	Manufacturer's name and address			
	Type			
	Size			
	Type of operation			
(m)	Lighting fittings – Fluorescent			

Information Copy

No	Item	Units	Required	Tendered
	Manufacturer's name and address			
	Type			
	Rating			
	Harmonic content			
(n)	Lighting fittings - emergency			
	Manufacturer's name and address			
	Type			
	Rating	W		
(o)	Switch yard lighting			
	Manufacturer's name and address			
	Type			
	Rating	W		

Information Copy

3 TIMES FOR DELIVERY & COMPLETION AND CONTRACT COMPLETION TIMES

3.1 TIMES FOR DELIVERY AND COMPLETION

The individual dates are all contractually binding.

The Contractor shall complete the delivery, installation, integration, testing, and commissioning of all systems, hardware, software, and associated facilities strictly in accordance with the approved implementation schedule. The completion times shall include all activities necessary for the fully integrated and operational condition of the SCADA/EMS core system, database clusters, communication gateways, operator workstations, video wall systems, Renewable Energy Management System (REMS) modules, cybersecurity infrastructure, auxiliary control panels, networking equipment, and all associated civil, electrical, and mechanical works. Each subsystem shall be delivered and commissioned as a fully functional unit, ensuring that subsequent shutdowns or interruptions to NSCC operations are unnecessary or restricted only to short, planned outages approved by the Employer.

The dates assume and order is placed by Week No. 1. Key to dates be provided as follows.

E - Target completion dates planned by NSO

D - Implementation (delivery, erection and commissioning) date guaranteed by contractor.

C - Shipping completion dates guaranteed by contractor.

B - Date of arrival of first shipment guaranteed by contractor.

A - Earliest date by which access is required by the contractor.

Information Copy

Site	A Earliest Access Permitted Week No.	B Week No.	C Week No.	D Week No.	E Week No.
SCADA/EMS/REMS Modernization Project • SCADA/EMS Core Servers and Virtualization Cluster • Database and Historian Servers • Operator Workstations and Consoles • Control Center Operations Display System and Visualization Platform • REMS Integration Platform and APIs • Cybersecurity and Network Segmentation Infrastructure • Backup NSCC System Deployment • Civil, Electrical and Mechanical Modifications • Final Testing, Training and System Handover					

Notes:

1. The above schedule shall be finalized jointly by the Employer and Contractor during the project kick-off meeting.

2. All milestone dates shall be reflected in the Contractor's approved project management system.

3. Any delay attributable to the Contractor shall be governed by the provisions of the Contract under 'Delays and Liquidated Damages.'

4. Partial or staged commissioning shall only be permitted with the prior written approval of the Employer.

5. The Employer reserves the right to adjust the milestone sequence based on site readiness or interdependent project phases.

Information Copy

3.2 CONTRACT COMPLETION TIMES

The times entered below shall be those used to calculate the completion dates for the various sections of the Contract together with the overall Time for completion as referred to in Clauses and of the Conditions of Contract.

	Completion Time in Calendar months calculated from the date of Commencement
Completion Time for the Contract	- …….. Months.
Completion Time for Works at NSCC	- ……Months
Completion Time for Works at Remote Ends	- ……Months
Completion Time for Trainings	- ……Months.

Information Copy

4 DEVIATIONS FROM SPECIFICATION

(To be completed by the Bidder).

Any details that will lead to deductions of final Bid price shall not be inserted.

The bidder shall list below all deviations of his bid to the commercial and technical bidding conditions / specifications of these bidding documents.

It shall not be necessary for the employer to examine the standard literature and documents of the manufacturer to determine the existence and extent of any exceptions or deviations from this specification.

No other deviation stated elsewhere in the bid would be considered as valid deviations during the contract period.

Volume	Clause No.	Proposed Deviations

Information Copy

5 MANUFACTURES' AND SUBCONTRACTORS' STATEMENT OF EXPERIENCE

5.1 TENDERER'S STATEMENT OF PREVIOUS EXPERIENCE

The Tenderer shall complete this schedule giving full details of SCADA, EMS, DMS, or similar Control Centre projects of comparable scope, complexity, and technology that have been completed or are in the course of completion by the Tenderer within the last ten (10) years.

Each project listed shall include essential particulars such as the scope of work, system configuration, implemented technologies, contract value, client, and consultant.

Failure to complete this schedule with complete and verifiable details, or to provide the requested supporting documentation (e.g., completion certificates, client references, or performance confirmations), may render the offer liable to rejection.

The Tenderer shall also provide details of the manufacturers and major subcontractors proposed under this Contract, together with their experience in the design, manufacture, and supply of systems of similar nature and capacity in a separate sheet if required.

Name of the Project					
Contract No.					
Country					
System Type and Platform					
Scope of Works					
Purchaser					
Consultant or Employer					
No. of Grid Substations and Power Plants Integrated					
Contract Award Date					

Information Copy

Contractual Completion Date					
Actual Completion Date					
Contract Value					

Information Copy

5.2 KEY PERSONNEL

Designation	Name of (i) Nominee (ii) Alternate	Summary of Qualification Experience and Present Occupation	Year of Birth
<u>Headquarters</u>			
Project Director			
Project Manager			
Engineering Design Staff			
Other Key Staff (Give Designation)			
<u>Site Office</u>			
Project Manager			
Site Manager			
Deputy Site Manager			
Supervising Engineers			
Implementation Supervisors			
Other Key Staff			

The Tenderer shall list in this Schedule the key personnel (including first nominee and the second choice alternate) he will employ from headquarters and from site office to direct and execute the works.

Information Copy

5.3 CONTRACTOR'S SITE PERSONNEL

Implementation Staff

The contractor shall give below the status and numbers of staff required for implementation of the project and the estimated period for which they will be retained on site.

Supervisory and expatriate staff:	
(a) Bachelor status	
(b) Married status	

Position	Months
<u>Headquarters</u> Project Director Project Manager Other Key Staff (Give Designation) <u>Project Office in Sri Lanka</u> Project Manager <u>Site Office</u> Site Manager Deputy Site Manager Supervising Engineers Construction Supervisors Other Key Staff	

Information Copy

5.4 SUB-CONTRACTORS

Item	Element of work	Approximate value	Name and address of Sub	Statement of Similar Works

The tenderer shall enter in this schedule a list of the sections and appropriate value of the work for which the purposes to use sub-contractors, together with the names and addresses of the proposed sub-contractors. The tenderer shall also enter a statement of similar works

Previously executed by the proposed sub-contractors, including description, location and value of works, year completed, and name and addresses of the Employer Notwithstanding such information the tenderer, if awarded the contract, shall remain entirely and solely responsible for the satisfactory completion of the Works.

Information Copy

6 DOCUMENTS, DRAWINGS AND INFORMATION TO BE SUBMITTED WITH THE TECHNICAL BID

The following documents & drawings shall be submitted with the Technical BID.

6.1 Duly signed Part A – Scope of Works in Section VI – Employer's Requirement in Part 2 – Employer's Requirements in Volume 4 of 8.

6.2 Duly signed Part B – Technical Specification in Section VI – Employer's Requirement in Part 2 – Employer's Requirements in Volume 5 of 8.

6.3 Duly signed Part C – Drawings in Section VI – Employer's Requirement in Part 2 – Employer's Requirements in Volume 6 of 8.

6.4 Duly completed and signed Supplementary Information,

1. Manufacturers and place of manufacture and testing.
2. Technical particulars and guarantees.
3. Times for delivery and completion and contract completion.
4. Deviations from the Specification.
5. Manufacturers' and subcontractors' statement of experience.
6. Documents, drawings and information to be submitted.
7. Confirmation of Adherence to the Environmental Acts, Regulation and/or Guidelines.

6.5 Duly signed Section VII - General Conditions in Part 3 – Conditions of Contract and Contract Forms in Volume 8 of 8.

6.6 Duly signed Section VIII - Particular Conditions in Part 3 – Conditions of Contract and Contract Forms in Volume 8 of 8.

6.7 Duly signed Section IX –Contract Forms in Part 3 – Conditions of Contract and Contract Forms in Volume 8 of 8.

6.8 All documents required to demonstrate bidder's ability and capability as specified in Section 3 – Evaluation and Qualification Criteria in Part I – Bidding Procedures in Volume I of 8.

This shall include but not limited to, verifiable evidence to prove that Bidder meets the qualification requirements in Item 2 of Section 3 – Evaluation and Qualification Criteria.

6.9 Verifiable evidence of manufacturers' experience in manufacturing comparable type of equipment offered under this bid to meet the criteria of minimum of ten (10) years successful experience in manufacturing comparable equipment.

Information Copy

6.10 Verifiable evidence of service experience of equipment offered under this bid to meet the criteria of minimum of ten (10) years successful experience in manufacturing comparable equipment.

6.11 Irrespective of the fact that the Bidder has a local agent or sub-agent, representative or nominee to act for or on behalf of the Bidder, the Bidder shall register himself with an appropriate authority (registrar) under Public Contract Act. No. 03 of 1987 and a copy of the certificate issued by the registrar shall be submitted along with the Bid. Registration Certificate shall be included in the bids without which they shall be rejected.

6.12 Certificates issued by an independent international organization to ensure compliance with the ISO 9001:2000 standards by the Bidder and Manufacturers of all main equipment listed below,

1. Servers
2. Operator Consoles and Workstations
3. Control Center Operations Display System
4. Local Area Network (LAN) - Switches, Routers, & Cabling
5. Storage Area Network (SAN)
6. GPS Time Server
7. Cyber Security Appliances
8. Power Supply Units
9. Patch Panels and Other Cabinets
10. EMS and REMS Applications, Module and Software Components

6.13 Authorization letters shall be provided with the bid in respect of following items not manufactured by the bidder,

1. Servers
2. Operator Consoles and Workstations
3. Control Center Operations Display System
4. Local Area Network (LAN) - Switches, Routers, & Cabling
5. Storage Area Network (SAN)
6. GPS Time Server
7. Cyber Security Appliances
8. Power Supply Units
9. Patch Panels and Other Cabinets
10. EMS and REMS Applications, Module and Software Components

6.14 The manufacturer's guarantee that they have an established department that will serve the National System Operator (Pvt) Ltd in supply of spares for at least 10 years for all equipment listed in 6.13 above.

6.15 The manufacturer's guarantee that they have an established department that will serve the National System Operator (Pvt) Ltd to provide advisory service

Information Copy

with regard to maintenance and overhauling at least for 10 years for all equipment listed in 6.13 above.

6.16 Layout drawings of NSCC Configuration and Contractor shall provide detailed migration plan for the migration of exiting system to new modernized SCADA/EMS/REMS system.

6.17 General bar chart of the design, manufacturing, shipping, erection and commissioning schedule considering the migration plan.

6.18 Type Test certificates in accordance with standards specified in relevant Chapters in Volume 5 of 8, issued by an independent laboratory or Type Tests witnessed by Employer for,

1. Servers
2. Operator Consoles and Workstations
3. Control Center Operations Display System
4. Local Area Network (LAN) - Switches, Routers, & Cabling
5. Storage Area Network (SAN)
6. GPS Time Server
7. Cyber Security Appliances
8. Power Supply Units
9. Patch Panels and Other Cabinets

(note: It the offered equipment is manufacture under license, type test certificate for the equipment for the manufacture in the offered factory shall be submitted with the bid.)

6.19 Parent company guarantee in case of joint ventures and subsidiaries.

6.20 Descriptive information for equipment being offered including;

1. List of recommended spare parts with prices.
2. List of special tools or fixtures required for installation, testing, maintaining and operating the equipment.
3. List of cost of special tools, lifting devices required for installation, operation and maintenance.

6.21 Details/drawings of panels and cabinets.

6.22 Typical arrangement drawings of control room and server room.

6.23 EMS related block diagrams as per Volume 6 of the Bidding document.

6.24 Typical diagrams of architecture of related EMS/SCADA/REMS and associated system similar to the architecture offered.

6.25 In case of alternative bids are submitted as per sub clause 13.3 of ITB,

6.26 Any other material required to be completed and submitted by bidders in accordance with the instruction to bidders.

Information Copy

7 ADHERENCE TO THE ENVIRONMENTAL ACTS, REGULATIONS AND / OR GUIDELINES

The Bidders shall submit with their Technical Bid, written confirmation certifying that they will comply with Environmental Safeguard Documents prepared under Environmental Assessment Guidelines of Asian Development Bank (ADB).

1. Power System Reliability Strengthening Project: Updated Initial Environmental Examination (https://www.adb.org/projects/documents/sri-51122-002-iee-0)
2. Power System Strengthening and Renewable Energy Integration Project: Updated Resettlement Plan (https://www.adb.org/projects/documents/sri-51122-002-rp-0)

Information Copy

Section 6 - Employer's Requirements
Part E - BANK GUARANTIES AND CERTIFICATES, VARIATIONS AND ADJUSTMENTS ORDERS

Information Copy

Information Copy

9. Certificates

9.1 Form of Completion Certificate

Contract: [. . . . .*insert name of contract and contract identification details*. . . . .]

Date: .....

Certificate No.: .....

To: [. . . . .*insert name and address of contractor*. . . . .]

Dear Ladies and/or Gentlemen,

Pursuant to GCC Clause 24 (Completion of the Facilities) of the General Conditions of the Contract entered into between yourselves and the Employer dated [. . . . .*insert date*. . . . .], relating to the [. . . .*brief description of the Facilities* . . . .], we hereby notify you that the following part(s) of the Facilities was (were) complete on the date specified below, and that, in accordance with the terms of the Contract, the Employer hereby takes over the said part(s) of the Facilities, together with the responsibility for care and custody and the risk of loss thereof on the date mentioned below.

1. Description of the Facilities or part thereof: [. . . .*description* . . . .]

2. Date of Completion: [. . . .*date* . . . .]

However, you are required to complete the outstanding items listed in the attachment hereto as soon as practicable.

This letter does not relieve you of your obligation to complete the execution of the Facilities in accordance with the Contract nor of your obligations during the Defect Liability Period.

Very truly yours,

[. . . .*Signature* . . . .]

Project Manager

Information Copy

9.2 Form of Operational Acceptance Certificate

Contract: [. . . . .*insert name of contract and contract identification details*. . . . .]

Date: .....

Certificate No.: .....

To: [. . . . .*insert name and address of contractor*. . . . .]

Pursuant to GCC Subclause 25.3 (Operational Acceptance) of the General Conditions of the Contract entered into between yourselves and the Employer dated [. . .*date*. . .], relating to the [. . .*brief description of the facilities*. . .], we hereby notify you that the Functional Guarantees of the following part(s) of the Facilities were satisfactorily attained on the date specified below.

1. Description of the Facilities or part thereof: [. . . *description* . . .]

2. Date of Operational Acceptance: [. . . *date* . . .]

This letter does not relieve you of your obligation to complete the execution of the Facilities in accordance with the Contract nor of your obligations during the Defect Liability Period.

Very truly yours,

[. . . .*Signature* . . . .]

Project Manager

Information Copy

10. Change Orders

10.1 Change Order Procedure
10.1.1 General
10.1.2 Change Order Log
10.1.3 References for Changes

10.2. Change Order Forms
10.2.1 Request for Change Proposal
10.2.2 Estimate for Change Proposal
10.2.3 Acceptance of Estimate
10.2.4 Change Proposal
10.2.5 Change Order
10.2.6 Pending Agreement Change Order
10.2.7 Application for Change Proposal

Information Copy

10.1. Change Order Procedure

10.1.1 General

This section provides samples of procedures and forms for implementing changes in the Facilities during the performance of the Contract in accordance with GCC Clause 39 (Change in the Facilities) of the General Conditions.

10.1.2 Change Order Log

The Contractor shall keep an up-to-date Change Order Log to show the current status of Requests for Change and Changes authorized or pending. Entries of the Changes in the Change Order Log shall be made to ensure that the log is up-to-date. The Contractor shall attach a copy of the current Change Order Log in the monthly progress report to be submitted to the Employer.

10.1.3 References for Changes

(1) Request for Change as referred to in GCC Clause 39 shall be serially numbered CR-X-nnn.

(2) Estimate for Change Proposal as referred to in GCC Clause 39 shall be serially numbered CN-X-nnn.

(3) Acceptance of Estimate as referred to in GCC Clause 39 shall be serially numbered CA-X-nnn.

(4) Change Proposal as referred to in GCC Clause 39 shall be serially numbered CP-X-nnn.

(5) Change Order as referred to in GCC Clause 39 shall be serially numbered CO-X-nnn.

Note:
(a) Requests for Change issued from the Employer's Home Office and the Site representatives of the Employer shall have the following respective references:

Home Office	CR-H-nnn
Site	CR-S-nnn

(b) The above number "nnn" is the same for Request for Change, Estimate for Change Proposal, Acceptance of Estimate, Change Proposal and Change Order.

Information Copy

10.2 Change Order Forms

10.2.1 Request for Change Proposal Form

[*Employer's letterhead*]

To: [*Contractor's name and address*] Date:

Attention: [*Name and title*]

Contract Name: [*Contract name*]
Contract Number: [*Contract number*]

Dear Ladies and/or Gentlemen:

With reference to the captioned Contract, you are requested to prepare and submit a Change Proposal for the Change noted below in accordance with the following instructions within [*number*] days of the date of this letter [or on or before (*date*)].

1. Title of Change: [*Title*]
2. Change Request No./Rev.: [*Number*]
3. Originator of Change:
   Employer: [Name]
   Contractor (by Application for Change Proposal No. [Number Refer to Annex 6.2.7])
4. Brief Description of Change: [*Description*]
5. Facilities and/or Item No. of equipment related to the requested Change: [*Description*]
6. Reference drawings and/or technical documents for the request of Change:
   Drawing No./Document No. *Description*
7. Detailed conditions or special requirements on the requested Change: [*Description*]
8. General Terms and Conditions:
   (a) Please submit your estimate showing what effect the requested Change will have on the Contract Price.
   (b) Your estimate shall include your claim for the additional time, if any, for completing the requested Change.
   (c) If you have any opinion that is critical to the adoption of the requested Change in connection with the conformability to the other provisions of the Contract or the safety of the Plant or Facilities, please inform us in your proposal of revised provisions.
   (d) Any increase or decrease in the work of the Contractor relating to the services of its personnel shall be calculated.
   (e) You shall not proceed with the execution of the work for the requested Change until we have accepted and confirmed the amount and nature in writing.

[*Employer's name*]
[*Signature*]
[*Name of signatory*]
[*Title of signatory*]

Information Copy

10.2.2 Estimate for Change Proposal Form

[*Contractor's letterhead*]

To: [*Employer's name and address*] Date:

Attention: [*Name and title*]

Contract Name: [*Contract name*]
Contract Number: [*Contract number*]

Dear Ladies and/or Gentlemen:

With reference to your Request for Change Proposal, we are pleased to notify you of the approximate cost to prepare the below-referenced Change Proposal in accordance with GCC Subclause 39.2.1 of the General Conditions. We acknowledge that your agreement to the cost of preparing the Change Proposal, in accordance with GCC Subclause 39.2.2, is required before estimating the cost for change work.

1. Title of Change: [*Title*]
2. Change Request No./Rev.: [*Number*]
3. Brief Description of Change: [*Description*]
4. Scheduled Impact of Change: [*Description*]
5. Cost for Preparation of Change Proposal: [*insert costs, which shall be in the currencies of the contract*]

(a)	Engineering		(Amount)
	(i)	Engineer _____ hours (hrs) x _ rate/hr =	_____
	(ii)	Draftsperson _____ hrs x _____ rate/hr =	_____
		Sub-total _____ hrs	_____
		Total Engineering Cost	_____
(b)	Other Cost		_____
	Total Cost (a) + (b)		_____

[*Contractor's name*]
[*Signature*]
[*Name of signatory*]
[*Title of signatory*]

Information Copy

10.2.3 Acceptance of Estimate Form

[*Employer's letterhead*]

To: [*Contractor's name and address*] Date:

Attention: *[Name and title]*

Contract Name: *[Contract name]*
Contract Number: *[Contract number]*

Dear Ladies and/or Gentlemen:

We hereby accept your Estimate for Change Proposal and agree that you should proceed with the preparation of the Change Proposal.

1. Title of Change: [*Title*]
2. Change Request No./Rev.: [*Request number/revision*]
3. Estimate for Change Proposal No./Rev.: [*Proposal number/revision*]
4. Acceptance of Estimate No./Rev.: [*Estimate number/revision*]
5. Brief Description of Change: [*Description*]
6. Other Terms and Conditions: In the event that we decide not to order the Change accepted, you shall be entitled to compensation for the cost of preparing the Change Proposal described in your Estimate for Change Proposal mentioned in para. 3 above in accordance with GCC Clause 39 of the General Conditions.

[*Employer's name*]
[*Signature*]
[*Name of signatory*]
[*Title of signatory*]

Information Copy

10.2.4 Change Proposal Form

[*Contractor's letterhead*]

To: [*Employer's name and address*] Date:

Attention: [*Name and title*]

Contract Name: [*Contract name*]
Contract Number: [*Contract number*]

Dear Ladies and/or Gentlemen:

In response to your Request for Change Proposal No. [Number], we hereby submit our proposal as follows:

1. Title of Change: [*Name*]
2. Change Proposal No./Rev.: [*Proposal number / revision*]
3. Originator of Change: Employer: [*Name*] / Contractor: [*Name*]
4. Brief Description of Change: [*Description*]
5. Reasons for Change: [*Reason*]
6. Facilities and/or Item No. of Equipment related to the requested Change: [*Facilities*]
7. Reference drawings and/or technical documents for the requested Change:
   [*Drawing/Document No./Description*]
8. Estimate of increase/decrease to the Contract Price resulting from the Change Proposal:

		Amount [*insert amounts in the currencies of the Contract*]
(a)	Direct material	_____
(b)	Major construction equipment	_____
(c)	Direct field labor (Total hrs)	_____
(d)	Subcontracts	_____
(e)	Indirect material and labor	_____
(f)	Site supervision	_____
(g)	Head office technical staff salaries	
	Process engineer _____ hrs @ _____ rate/hr	_____
	Project engineer _____ hrs @ _____ rate/hr	_____
	Equipment engineer _____ hrs @ _____ rate/hr	_____
	Procurement _____ hrs @ _____ rate/hr	_____
	Draftsperson _____ hrs @ _____ rate/hr	_____
	Total _____hrs	

Information Copy

(h) Extraordinary costs (computer, travel, etc.) _____

(i) Fee for general administration, % of Items _____

(j) Taxes and customs duties _____

Total lump sum cost of Change Proposal [*Sum of items (a) to (j)*]

Cost to prepare Estimate for Change Proposal [*Amount payable if Change is not accepted*]

9. Additional time for Completion required due to Change Proposal
10. Effect on the Functional Guarantees
11. Effect on the other terms and conditions of the Contract
12. Validity of this Proposal: within [Number] days after receipt of this Proposal by the Employer
13. Other terms and conditions of this Change Proposal:
    (a) You are requested to notify us of your acceptance, comments or rejection of this detailed Change Proposal within [Number] days from your receipt of this Proposal.
    (b) The amount of any increase and/or decrease shall be taken into account in the adjustment of the Contract Price.
    (c) Contractor's cost for preparation of this Change Proposal: [*. . . .insert amount. This cost shall be reimbursed by the employer in case of employer's withdrawal or rejection of this Change Proposal without default of the contractor in accordance with GCC Clause 39 of the General Conditions . . . .*]

[*Contractor's name*]
[*Signature*]
[*Name of signatory*]
[*Title of signatory*]

Information Copy

10.2.5 Change Order Form

[*Employer's letterhead*]

To: *[Contractor's name and address]* Date:

Attention: *[Name and title]*

Contract Name: *[Contract name]*
Contract Number: *[Contract number]*

Dear Ladies and/or Gentlemen:

We approve the Change Order for the work specified in the Change Proposal (No. [*number*]), and agree to adjust the Contract Price, Time for Completion, and/or other conditions of the Contract in accordance with GCC Clause 39 of the General Conditions.

1. Title of Change: [*Name*]
2. Change Request No./Rev.: [*Request number / revision*]
3. Change Order No./Rev.: [*Order number / revision*]
4. Originator of Change: Employer: [*Name*] / Contractor: [*Name*]
5. Authorized Price:
   Ref. No.: [*Number*] Date: [*Date*]
   Foreign currency portion [*Amount*] plus Local currency portion [*Amount*]
6. Adjustment of Time for Completion
   None Increase [*Number*] days Decrease [*Number*] days
7. Other effects, if any

Authorized by: _____ Date: _____
Employer

Accepted by: _____ Date: _____
Contractor

Information Copy

10.2.6 Pending Agreement Change Order Form

[*Employer's letterhead*]

To: [*Contractor's name and address*] Date:

Attention: [*Name and title*]

Contract Name: [*Contract name*]
Contract Number: [*Contract number*]

Dear Ladies and/or Gentlemen:

We instruct you to carry out the work in the Change Order detailed below in accordance with GCC Clause 39 of the General Conditions.

1. Title of Change: [*Name*]
2. Employer's Request for Change Proposal No./Rev.: [*number/revision*] dated: [*date*]
3. Contractor's Change Proposal No./Rev.: [*number / revision*] dated: [*date*]
4. Brief Description of Change: [*Description*]
5. Facilities and/or Item No. of equipment related to the requested Change: [*Facilities*]
6. Reference Drawings and/or technical documents for the requested Change:
   [*Drawing / Document No. / Description*]
7. Adjustment of Time for Completion:
8. Other change in the Contract terms:
9. Other terms and conditions:

[*Employer's name*]

[*Signature*]

[*Name of signatory*]

[*Title of signatory*]

Information Copy

10.2.7 Application for Change Proposal Form

[*Contractor's letterhead*]

To: [*Employer's name and address*] Date:

Attention: [*Name and title*]

Contract Name: [*Contract name*]
Contract Number: [*Contract number*]

Dear Ladies and/or Gentlemen:

We hereby propose that the work mentioned below be treated as a Change in the Facilities.

1. Title of Change: [*Name*]
2. Application for Change Proposal No./Rev.: [*Number / revision*]dated: [*Date*]
3. Brief Description of Change: [*Description*]
4. Reasons for Change:
5. Order of Magnitude Estimation (amount in the currencies of the Contract): [*Amount*]
6. Scheduled Impact of Change:
7. Effect on Functional Guarantees, if any:
8. Appendix:

[*Contractor's name*]

[*Signature*]

[*Name of signatory*]

[*Title of signatory*]

Information Copy

11. Personnel Requirements

Using Form PER - 1 and PER - 2 in Section 4 (Bidding Forms), the Bidder must demonstrate that it has personnel who meet the following requirements:

No.	Position	Total Work Experience [years]	Experience In Similar Work [years]
1	Project Manager	10	5
2	Design Engineers	10	5
3	System Engineers	5	5
4	Implementation Supervisors	5	5
5	Safety Officers	5	5

Using Form EXP-6 in Section 4 (Bidding Forms), the Bidder must demonstrate that it has EHS personnel who meet the following requirements:

Key Personnel as determined by the EMP and other safeguard management plans (Bidder shall fill below table)

Item No.	Position/specialization	Relevant academic qualifications	Minimum years of relevant work experience	Minimum time on-site (%FTE)
1				
2				
3				

Information Copy

12. Equipment Requirements

Using Form EQU in Section 4 (Bidding Forms), the Bidder must demonstrate that it has the key equipment (Bidder shall fill below table):

No.	Equipment Type and Characteristics	Minimum Number Required

Information Copy